

The Long-Run Elasticity of Demand for Commuter Rail Transportation

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I. INTRODUCTION

The divergence in short-run and long-run elasticities can help explain the familiar cycle of falling public transit ridership, fare increases, and service elimination. Since the total ridership impact of a change in fares or service quality may not be realized in the current period, a transit authority facing a single-year budget constraint may increase price or reduce service to balance the budget in the current year, but these changes may result in larger deficits in subsequent periods.

Despite its relevance for transportation policy, no estimates of the long-run elasticity of demand for public transportation have been published. The commonly accepted elasticity estimates based on the discrete choice framework pioneered by McFadden [12, 13] reflect only changes in the modal split of travel for a fixed population and hence are short run.¹ Long-run elasticity estimates, however, should also account for the indirect ridership impacts of transportation prices and quality through their effects on the size, distribution, and transportation spending of the population.²

This paper measures the long-run change in ridership induced by price and service changes using a new location-specific panel data set. Rather than analyze the effect of transportation prices and service attributes on *individuals'* modal choices at any point in time, this analysis examines the impacts of prices and service attributes on ridership at *locations* over time. We estimate a dynamic, fixed-effects model which captures both long-term

¹Winston [16] notes that there are three components of total demand elasticity: "mode divergence," "destination divergence," and "trip generation." He claims that mode divergence is the largest component, although he does not cite any evidence in support of this claim.

²Anas [1], Anas and Chu [2], Thobani [14], and Ben-Akiva and Watanatada [5] have documented the importance of transportation prices and quality in the individual's residence location and transportation decisions.

and short-term responses to price and service changes, controlling for initial demographic differences across locations.

A panel data set on the Southeastern Pennsylvania Transportation Authority (SEPTA) commuter rail system was assembled to address the issue of the long-run impacts of price and service changes. This data set contains information on ridership, fares, and service attributes for 129 stations on the SEPTA system for 12 observation points during the period 1978–1986. Based on these data we find that ridership is more than twice as elastic in the long run as in the short run. Ridership on SEPTA, which is price inelastic in the short run, is price elastic in the long run. The characteristics of service such as frequency and speed of trains, and alternative transportation prices, have significant effects on ridership, which are substantially larger in the long run than in the short run.

The findings suggest that reductions in public transportation subsidies that result in higher fares and lower service quality may produce higher subsidy costs per rider than would be the case with higher total subsidy.³ Thus, the results from this analysis support the common public perception that raising public transit fares and reducing service simply reduce ridership, requiring further fare increases and service cuts.

II. THE EVOLUTION OF COMMUTER RAIL DEMAND OVER TIME

At a given location, commuter rail ridership depends on the share of potential riders choosing the train and the number of potential riders. In the short run, the pool of potential riders is fixed; consumer response to changes in price and service levels is limited to changes in the modal choice or number of trips. In the long run, changes in prices and service attributes affect the number of potential riders through their impacts on choice of residence, job location, and spending on private transportation.

If the pricing and quality of train service make it an attractive alternative, people and firms make location and investment decisions that are consistent with high levels of future ridership. Individuals who work along the train lines sort themselves into residential locations that have train service.⁴ Land in areas with service may become more valuable as riders bid up property values.⁵ People who work at locations not well served by the commuter system choose residential locations that are not near train stations to avoid paying higher housing prices. This sorting behavior

³Increased subsidies are not necessarily translated into lower fares or improved service. See J. Gomez-Ibanez [7].

⁴Although little evidence is available on the degree of sorting, mobility is sufficient for this effect to be large. About 40% of all Americans moved between 1980 and 1985 (source: "The Statistical Abstract of the United States, 1987," Table 29 [15]).

⁵See, for example, Damm *et al.* [6].

results in locations with good train service having a large number of people travelling to destinations that are accessible by train.

However, if train service is of poor quality, is too expensive, or if future service is uncertain, the possibility of future train service will not weigh heavily in the investment and location decisions of individuals or firms. The resultant spatial distribution of people and jobs, as well as the level of investment in private transportation, is not likely to result in high future ridership.

III. MODEL

An individual's demand for rides on the public transit system depends on the price of a ride p_0 , the price of alternatives p_a , service attributes A_0 , and attributes of alternatives A_a , as well as on the individual's destinations D , transportation capital stock K , and income Y . Individuals' destinations affect demand through their effect on the prices of both public and private transportation. Destinations that are not served by public transportation have an infinite price for public transportation. The price of private transportation alternatives depends on the individual's private transportation capital stock. If an individual does not own a car, for example, the price of alternative transportation is likely to be high.⁶ Thus, the demand for rail trips R , by individual i near station j during period t is

$$R^{ijt} = h(p_0^{jt}(D^{it}), p_a^{jt}(D^{it}, K^{it}), Y^{it}, A_0^{jt}(D^{it}), A_a^{jt}(D^{it}, K^{it})). \quad (1)$$

The total demand for rides from j during t is the sum of all the demands of individuals near j :

$$\begin{aligned} R^{jt} &= \sum_{i=1}^{N^j} R^{ijt} \\ &= H(p_0^{jt}, p_a^{jt}, A_0^{jt}, A_a^{jt}, \phi^{jt}(D^{jt}, K^{jt}, Y^{jt}, N^{jt})). \end{aligned} \quad (2)$$

The parameter ϕ^{jt} represents the size of the potential market at j in t . It depends on the distribution of destinations, the private transportation capital stock, and the income of the people in j , as well as the number of people in j at t , N^{jt} . The share of the potential market using the train is determined by p_0 , p_a , A_0 , and A_a . Ridership at j is a function H of the determinants of market share and the size of the market.

⁶Fixed costs of automobile ownership and transactions costs of leasing a car are likely to cause effective prices of the alternative to be different for auto owners and nonowners in the short term.

At any t , the determinants of ϕ^j are constants, but D , K , Y , and N^j may change over time, in part as a function of the price and service attributes of the transportation system. The potential market for the train changes as individuals make long-term choices. Assuming that these choices are made gradually and incur transaction costs, decisions made in prior periods affect the size and composition of the current population in a particular location. Factors affecting previous decisions have current effects on the market.

By partitioning factors affecting the determinants of ϕ into transportation-related factors and nontransportation-related factors, one can postulate a set of functions that determines D , K , Y , and N^j :

$$\begin{aligned} D^{jt} &= D(p_0^{j,t-i}, p_a^{j,t-i}, A_0^{j,t-i}, A_a^{j,t-i}; X^{j,t-i}) & i = 1, 2, \dots, \infty \\ K^{jt} &= K(p_0^{j,t-i}, p_a^{j,t-i}, A_0^{j,t-i}, A_a^{j,t-i}; X^{j,t-i}) & i = 1, 2, \dots, \infty \\ Y^{jt} &= Y(p_0^{j,t-i}, p_a^{j,t-i}, A_0^{j,t-i}, A_a^{j,t-i}; X^{j,t-i}) & i = 1, 2, \dots, \infty \\ N^{jt} &= N(p_0^{j,t-i}, p_a^{j,t-i}, A_0^{j,t-i}, A_a^{j,t-i}; X^{j,t-i}) & i = 1, 2, \dots, \infty, \end{aligned} \quad (3)$$

where D , K , Y , and N are all functions relating current and past prices, attributes, and other factors to D^{jt} , K^{jt} , Y^{jt} , and N^{jt} ;

$X^{j,t-i}$ = factors affecting D^{jt} , K^{jt} , Y^{jt} , and N^{jt} , which are uncorrelated with transportation system prices or attributes.

Given (3), ϕ^{jt} can be expressed as a function of current and past values of p , A , and X :

$$\phi^{jt} = \Psi(p_0^{j,t-i}, p_a^{j,t-i}, A_0^{j,t-i}, A_a^{j,t-i}; X^{j,t-i}) \quad i = 1, 2, \dots, \infty. \quad (4)$$

Equation (4), however, is not observable, but by substituting (4) into (2), we obtain an equation for ridership with only observable variables

$$R^{jt} = h(p_0^{j,t-i}, p_a^{j,t-i}, A_0^{j,t-i}, A_a^{j,t-i}; X^{j,t-i}) \quad i = 0, 1, \dots, \infty. \quad (5)$$

Ridership is a function of current and lagged price, service attributes and exogenous factors unrelated to previous price and service levels.

One concern in this framework is whether transportation prices and attributes are simultaneously determined with ridership, which would make estimation of (5) difficult. Fortunately, the nature of decision making

on SEPTA assures that prices and service attributes are set independently of current ridership. The frequency of trains cannot be simultaneously determined with current period ridership as schedules are changed half-yearly and the timing of price changes depends on the pace of the public hearing process, which is required for all fare changes. Supply is generally equilibrated to demand by changing the number of cars on a train rather than changing frequency of trains or price, at least in the short run. Other attributes such as speed are functions of the infrastructure and are not determined simultaneously with ridership. As a practical matter, price and service attributes are probably more a function of changes in subsidy level and operating costs than of changes in ridership.

IV. SPECIFICATION AND ESTIMATION

Equation (5) can be specified as a dynamic fixed-effects model. The fixed effects control for the demographic differences that determine ϕ^j , the size of the potential market at each station.⁷ To limit the number of parameters we assume that the effects of lagged variables decline geometrically over time and that the decay rate is the same for each explanatory variable and each location. This assumption implies that the greatest impact occurs in the initial period and that the ratio of short run to total impacts is the same for all lagged variables for all locations. Finally, we assume that the attributes of alternatives, except for price, are unchanged during the sample period.

Given these assumptions,

$$R^{jt} = \beta_1 \sum_{k=0}^{\infty} \lambda^k p_0^{j,t-k} + \beta_2 \sum_{k=0}^{\infty} \lambda^k p_a^{j,t-k} + \beta_3 \sum_{k=0}^{\infty} \lambda^k A_0^{j,t-k} + \delta S + e^{jt} \quad (6)$$

the model can be estimated, where λ is the geometric lag parameter, $0 \leq \lambda < 1$, S is a vector of dummy variables for location, and δ is a vector of parameters that corresponds to the fixed effects. Taking differences from within-group means, (6) can be rewritten as

$$\tilde{R}^{jt} = \beta_1 \sum_{k=0}^{\infty} \lambda^k \tilde{p}_0^{j,t-k} + \beta_2 \sum_{k=0}^{\infty} \lambda^k \tilde{p}_a^{j,t-k} + \beta_3 \sum_{k=0}^{\infty} \lambda^k \tilde{A}_0^{j,t-k} + e^{jt}, \quad (7)$$

where $\tilde{}$ denotes differences from within-group means.⁸

⁷The estimates for a cross-section regression may be misleading because there may be demographic differences across locations that in part are the result of presample transportation policies but are correlated with current period variables.

⁸Within-group means are calculated as the mean for each station over all periods for which data are available, so for each successive lagged value of an independent variable the within-group mean is based on fewer observations than the preceding mean.

The β 's give the immediate response to a change in price and service variables. This impact includes changes in modal choice as well as current period changes in location or investment in private transportation, which affect ridership. The total impact of an independent variable is $\beta/(1 - \lambda)$; the difference between the short run impact and the total impact reflects long-run adjustments through relocation and transportation investments. The mean lag $\lambda/(1 - \lambda)$ is the number of periods until half of the total impact has occurred.

If e is serially uncorrelated, (7) can be estimated directly by adapting a maximum likelihood approach suggested by Klein [10] for panel data as in Hill [8] and Baltegi and Levin [4].⁹ If e is assumed to be AR1, the Zellner and Geisel (Z-G) [17] method can be applied.¹⁰ Both of these procedures involve writing (7) as a nonlinear regression model consisting of an observable component and a truncation parameter that is a function of λ .

Equation (7) can be transformed by successive substitution to

$$\tilde{R}^{jt} = \lambda \tilde{R}^{j,t-1} + \beta_1 \tilde{p}_0^{jt} + \beta_2 \tilde{p}_a^{jt} + \beta_3 \tilde{A}_0^{jt} + e + \lambda e^{t-1}. \quad (8)$$

Equation (8) can be estimated using instrumental variables as suggested by Levitan [11].¹¹ The instrumental estimator requires no assumptions about the distribution of e , but is inefficient.

V. DATA

A new panel data set on ridership, fares, and service attributes by station on SEPTA was assembled to estimate long-run price and service elasticities. The data set covers 129 of the 165 stations served by SEPTA

⁹In the case of panel data, however, the consistency of the estimates of β and λ depends on the assumptions concerning the initial conditions; see Anderson and Hsiao [3].

¹⁰Kiefer [9] discussed some peculiarities of estimating fixed-effects models with serial correlation. In particular, only the covariance structure of the transformed difference from means model is estimated. Still, these estimates are those necessary to calculate the appropriate standard errors of λ and β . In the dynamic model (7), the estimations are nonlinear in λ and β but additive in the fixed effects so the estimate of the autocorrelation parameter, ρ , is consistent and the subsequent data transformations in the Zellner-Geisel procedure yield consistent estimates of λ and β .

¹¹OLS estimation of (8) is likely to yield inconsistent estimates because the error term is correlated with the lagged dependent variable. An instrument can be constructed for the lagged dependent variable using the regression

$$\tilde{R}^{j,t-1} = \pi_0 \tilde{Z}^{j,t} + \pi_1 \tilde{Z}^{j,t-1} + \varepsilon.$$

The predicted value of $\tilde{R}^{j,t-1}$ from $\hat{\pi}_0$ and $\hat{\pi}_1$ is then used in (8).

TABLE 1
Means by Observation Period

Period	Variables						
	Riders per station	Peak trains	Off- peak trains	Speed (MPH)	Round- trip price	Auto cost (variable)	Auto cost (fixed)
Fall 1978	404	7.9	20.0	24.3	3.30	6.25	4.50
Spring 1979	411	7.8	20.0	23.3	2.50	6.30	4.96
Fall 1979	415	7.9	20.1	23.3	2.50	6.30	4.96
Spring 1980	421	7.9	20.1	23.1	2.80	6.78	5.57
Fall 1980	387	7.9	20.2	23.2	3.60	6.78	5.57
Spring 1981	363	7.8	19.9	23.8	4.10	6.93	6.51
Fall 1981	327	6.6	14.7	24.2	4.50	6.93	6.51
Spring 1982	313	6.6	14.6	24.1	4.50	7.09	6.57
Fall 1982	305	6.6	14.6	24.3	4.50	7.09	6.57
Spring 1984	225	7.4	17.1	24.2	4.46	7.35	6.43
Fall 1985	287	7.6	18.4	21.5	4.86	7.80	6.69
Fall 1986	333	7.6	18.9	22.0	5.44	8.13	7.11
Average	349	7.5	18.2	23.4	3.80	7.00	6.00

for 12 observation periods between 1978 and 1986.¹² In addition, automobile operating, ownership, and parking costs have been added to the SEPTA data.

The data on ridership are counts of inbound boardings to the center city for a typical weekday. No breakdown by time of day is available at the station level. Pricing data have been collected for peak and off-peak travel. The average price per ride is based on the peak and off-peak fares weighted by the proportions of riders traveling peak, 0.68, and off-peak, 0.32, systemwide. Data on service attributes include the number of peak and off-peak trains, and the average speed of the trip.

The automobile price data consist of the variable cost of the commute, including parking, and the fixed cost of owning a car. The operating cost per mile and fixed cost per day are based on a new midsize car.¹³ The automobile operating costs per mile are multiplied by the number of miles from each station to the downtown destination and then multiplied by 2 to obtain the round-trip automobile operating costs from each station. Data on parking costs have been obtained for municipally owned parking

¹²Only stations for which data were available for the entire 12 observation periods were used. Six stations that experienced extreme changes in the competing transit system were also eliminated; this deletion has little effect on the results.

¹³Data are from various publications of "Your Driving Costs" from the AAA, summarized in "Motor Vehicle Facts and Figures, 1986" and "Your Driving Costs," 1986 edition.

TABLE 2
Percentage Changes in Variables 1978–1986

Variable	Zone 1	Zone 2	Zone 3	Zone 4	Zone 5
Riders per station	-41.0	-32.2	-6.5	-6.0	5.7
Number of peak trains	1.4	-15.3	1.6	5.3	12.1
Number of off-peak trains	-20.3	-21.3	3.5	15.1	32.0
Speed	-23.8	-11.7	-5.0	-6.1	-11.6
Round-trip fare	151.7	176.5	128.0	126.0	97.3
Variable cost—Auto	33.0	31.6	30.2	29.0	27.5
Fixed cost—Auto	57.8	57.8	57.8	57.8	57.8

facilities in center city Philadelphia.¹⁴ The total variable costs associated with commuting by car are the sum of the operating costs and the parking costs.

Table 1 summarizes changes in ridership, prices, and service attributes overtime. Between 1978 and 1980, average ridership per station on a given day rose slightly from an average of 402 riders per station to the sample peak of 421 in the spring of 1980. After large fare increases and a 20% reduction in service frequency, ridership fell over 25% by the end of 1982. In 1983, SEPTA took over operation of the system from CONRAIL and, in an effort to reduce costs, endured a strike that lasted over 3 months. The first ridership data available after the strike are for Spring 1984. By this point, ridership had fallen dramatically to only 225 riders per station. After 1984, ridership rose to 333 riders per station by the Fall of 1986.

Significant differences over time are revealed when the data are broken down by fare zone as Table 2 shows.¹⁵ Zones 1 and 2, which are the closest to the center city, have had declines in ridership of 41 and 32%, respectively. Ridership fell by only 6% in zones 3 and 4 and increased by about 6% in zone 5. The fall in ridership in the interior zones was accompanied by reductions in the total number of trains of 15 and 19% and nominal price increases of 152 and 177% for zones 1 and 2, respectively. In zone 5, where ridership increased, the total number of trains increased by 25% and the nominal price increase was only 97%. It appears that much of the

¹⁴These lots may not reflect parking costs on average because they are operated by an agency of the city government that may not pursue the same objectives as private operators. Still, the changes in municipal parking prices probably are correlated with changes in the private market and municipal lots represent a substantial part of the total capacity. Monthly parking passes are not taken into account.

¹⁵Since 1983, SEPTA commuter rail fares have been based on a zone system. The contours of the zone system are sufficiently irregular that it only roughly relates to mileage. During much of the sample period there is considerable price variation within zones since the zone-based fare system was not used until 1983.

TABLE 3
Estimation Results^a

Dependent variable: Ridership						
Dependent variable mean: 349.0						
Coefficient	M.L.					
	Klein		Z-G		I.V.	
	Coef	S.E.	Coef	S.E.	Coef	S.E. ^b
Number of peak trains	5.83	1.58	4.17	2.02	3.19	2.45
Number of Off-peak trains	3.54	0.59	4.19	0.74	4.88	0.96
Speed	3.67	1.01	2.47	1.27	1.68	1.48
Round-trip fare	-42.23	5.33	-37.67	6.25	-34.49	7.04
Variable cost—Auto	46.73	8.01	43.02	9.69	35.70	12.01
Fixed cost—Auto	14.63	6.43	20.43	8.79	9.40	9.49
Dummy, 1984	-84.07	8.70	-91.61	8.22	88.31	8.18
Dummy, 1985	-34.74	8.98	-10.88	9.01	10.81	11.58
λ	0.62	0.041	0.61	0.050	0.54	0.10
ρ			0.38	0.03		
Truncation parameter ^c	66.64	14.04	111.57	34.27		
Number of observations	1548		1419		1419	

^aEstimates of the fixed effects are not shown.

^bStandard errors are adjusted for using an instrumental variable.

^cThe truncation parameter is not estimated consistently.

ridership loss in zones 1 and 2 is a result of price increases and service reductions. Service improvements and a more moderate rate of fare increase in the more distant zones appear to have had a positive effect on ridership.¹⁶

VI. ESTIMATION RESULTS

Maximum likelihood estimates of (7) using the Klein procedure and the Z-G procedure are shown in the first two columns of Table 3. Column three shows the instrumental variables (IV) estimation. The Klein and Z-G estimations yield similar parameter estimates even though the serial correlation parameter in the Z-G estimation, ρ , is significantly different

¹⁶While some of the ridership loss in the close-in zones, especially zone 1, may have been caused by population declines in the city of Philadelphia (unrelated to the changes in the transportation system), these declines are small relative to the magnitude of the decrease in ridership. There has been, however, significant improvement in the quality of the bus and subway system in these zones over this period. In the more distant zones, population growth should have provided natural growth in ridership for the commuter rail system. However, employment has been booming in these outlying areas and much of the population now also works in those areas.

TABLE 4
Elasticities^a

Variable	Short run	Long run
Round-trip fare	-0.62	-1.59
Variable cost—Auto	1.05	2.69
Fixed cost—Auto	0.44	1.13
Number of peak trains	0.14	0.36
Number of off-peak trains	0.74	1.89
Speed	0.16	0.42

^aElasticities are evaluated at 1986 levels.

than zero. The IV estimates are also similar to the maximum likelihood estimates, although the standard errors are larger and the estimate of λ is somewhat smaller. Since the parameter estimates are similar in all three estimation techniques, the discussion that follows focuses only on the Z-G estimates.

The highly significant estimate of λ in the Z-G estimation of 0.61 indicates that the long-run response to changes in prices and service attributes are 2.6 times larger than short-run responses. The mean lag is 1.6 periods. Since the average duration between observations was about 8 months, this implies that chronologically, the mean lag is about 1 year.¹⁷

The coefficients on the price variables—fares and costs of alternative transportation—are highly significant. The coefficient of -37.67 on own price implies a short-run elasticity of about -0.62; the coefficient of 43.02 on the variable cost of an auto trip implies an elasticity of about 1.05. (The short-run and long-run elasticities are summarized in Table 4.) The average, daily fixed cost of owning an automobile has a smaller, though significant, effect; the coefficient of 20.43 implies a short-run elasticity of 0.44. One expects the fixed cost of auto ownership to have a smaller effect since automobiles are used for trips with which the commuter system does not compete.

In the long run, demand is strikingly elastic with respect to own price (-1.59), the variable cost of an auto trip (2.69), and the fixed cost of auto ownership (1.13). One must exercise caution in making forecasts using these elasticities since they are based on a simple linear model and forecasting out of the sample price range may be misleading. Still, it is likely that demand is considerably more price elastic than previously

¹⁷The time between observations is irregular. To see if this had an impact on the estimated parameters, separate estimations using only eight evenly spaced cross sections were made. The results are very similar to those reported here. The estimated lag parameter is almost identical, implying a slightly longer mean lag length in years.

thought. For example, the estimates imply that for a \$0.25 increase in fares, revenues increase \$9500 per day immediately, but after a half-year, they fall to \$1990. By the end of the first full-year revenue is reduced by \$3193 and after 4 years is below the original level by over \$14,600, based on an initial ridership of 100,000 per day and 1986 fares. Reduction in the variable cost and fixed costs of an auto trip have similar effects on ridership over time.

The impacts of service attributes on ridership are considerable. The coefficient on peak trains of 4.17 indicates that ridership would increase by nearly 42 riders in the short run if a peak train were added on a line with 10 station stops. The short-run elasticity of peak ridership with respect to number of peak trains is 0.14. The long-run elasticity is 0.36. For number of off-peak trains, the coefficient of 4.19 indicates a short-run elasticity of off-peak ridership of 0.74 and a long-run elasticity of 1.89. Because the off-peak elasticity is greater than 1, ridership *per train* could be increased by increasing the frequency of service. The off-peak elasticities are considerably higher than peak elasticities even though the estimated coefficients are similar because the ridership per off-peak train is lower on average than that for peak hour trains.¹⁸ One would expect off-peak journeys to be more frequency-elastic than peak trips, since they probably are not constrained to a regular schedule like work trips. If service is infrequent, off-peak journeys of uncertain duration may be inconvenient to make by train.

The coefficient of 2.47 on average speed implies a short-run elasticity of 0.16 and a long-run elasticity of 0.42. The speed coefficient is quite large, considering that the average speed in 1986 was only 22 miles per hour, so that a 1 mile per hour average speed increase is a 4.5% speed change. If average speed could be increased by 10 miles an hour, short-run ridership would increase by 7.2% and long-run ridership would increase by 18.5%. Further, if the 10 mile per hour increase in speed allowed frequency of operation to increase by 30% with the same crews and equipment, the total impact of speed would be over 44% in the long run.

In addition to the price and service variables, all three estimations have dummy variables for the 1984 and 1985 observations. These observations followed the 3-month strike in the spring of 1983. The large significant coefficient on the dummy for 1984 indicates that the strike continued to have a major impact on ridership a year after it ended. The estimates of

¹⁸The elasticity of peak ridership with respect to number of peak trains is based on 1986 peak ridership of 226 riders per day and an average number of peak trains of 7.6 per day. The elasticity of off-peak ridership with respect to number of off-peak trains is based on 1986 off-peak ridership of 107 riders per day and an average number of off-peak trains of 18.9 per day.

other coefficients are not, however, driven by the anomaly of the strike; similar results are obtained for all of the coefficients when only prestrike data are used.

VII. CONCLUSION

This paper provides estimates of the long-run price and service elasticities of ridership on a public transportation system. Rather than focus on the modal choice decisions of a given population with fixed origins, destinations, and private transportation investments, a methodology is developed that allows the estimation of the total elasticities, which include long-run effects of transportation prices and service characteristics on ridership through their effects on location and investment decisions. This framework is empirically implemented as a dynamic fixed-effects model using a new panel data set on the SEPTA commuter rail system. The estimates indicate that ridership is more than twice as elastic in the long run than in the short run.

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